📘 Walmart Weekly Sales — Metric Dictionary Definitions • Formulas • Assumptions • Governance

Purpose
This page documents the business definitions, formulas, assumptions, and governance checks used across the Walmart Weekly Sales dashboard pages. Use it as the single source of truth for KPI interpretation.

Metric Dictionary — Walmart Weekly Sales (BigQuery + Streamlit)

Data grain and core tables

Primary fact grain: `store_id + dept_id + week_start_date`

Core fields: `Weekly_Sales`, `IsHoliday`, `Store`, `Dept`, `Date`

Store attributes: `Type`, `Size` (from `raw_stores.csv`)

Optional features: `Temperature`, `Fuel_Price`, `CPI`, `Unemployment`, `MarkDown1-5` (from `raw_features.csv`)

Key assumptions (important)

- "Week" is defined by the dataset's `Date` (weekly observation).
- "Holiday week" is `IsHoliday = TRUE` (may not cover all holiday effects).
- Dataset contains **sales**, not profit/margin; we use sales-based proxies for performance.

North Star

1) Total Weekly Sales ($)

- **Definition:** Total sales in a given week across all stores and departments.
- **Formula (SQL):** `SUM(Weekly_Sales)`
- **Default filters:** none
- **Grain:** week_start_date
- **Notes:** Use as the top KPI tile; always show trend.

Sales volume and mix

2) Sales by Store ($)

- **Definition:** Total weekly sales per store.
- **Formula:** `SUM(Weekly_Sales)` grouped by `Store`
- **Grain:** store_id + week_start_date
- **Notes:** Use for leaderboards and “top movers.”

3) Sales by Department ($)

- **Definition:** Total weekly sales per department.
- **Formula:** `SUM(Weekly_Sales)` grouped by `Dept`
- **Grain:** dept_id + week_start_date

4) Sales by Store Type ($)

- **Definition:** Total weekly sales by store type (A/B/C).
- **Formula:** `SUM(Weekly_Sales)` grouped by `Type`
- **Grain:** store_type + week_start_date
- **Dependencies:** join to stores table on `Store`

5) Sales per Store ($/store)

- **Definition:** Average weekly sales per active store.
- **Formula:** `SUM(Weekly_Sales) / COUNT(DISTINCT Store)`
- **Grain:** week_start_date (or segment + week)

- **Notes:** Use to compare productivity over time independent of store count.

6) Sales per Square Foot ($/sqft)

- **Definition:** Sales normalized by store size.
- **Formula:** `SUM(Weekly_Sales) / SUM(Size)` (within chosen group)
- **Grain:** week_start_date (or store_type + week)
- **Dependencies:** stores table has `Size`
- **Notes:** Works best by store type or region-like groupings; avoid dividing per dept-store rows unless carefully aggregated.

7) Department Contribution (% of Sales)

- **Definition:** Share of total sales attributable to each department within a period.
- **Formula:**
  - `dept_sales = SUM(Weekly_Sales)` by Dept
  - `total_sales = SUM(Weekly_Sales)` (same filters)
  - `dept_contribution = dept_sales / total_sales`
- **Grain:** dept_id + period
- **Notes:** Use in Driver page to show which depts drive overall results.

Growth

8) WoW Sales Growth (%)

- **Definition:** Week-over-week percent change in total sales.
- **Formula:** `(sales_this_week - sales_last_week) / sales_last_week`
- **Grain:** week_start_date
- **Implementation note:** Use `LAG(total_sales) OVER (ORDER BY week_start_date)`
- **Edge cases:** If last week sales = 0, return NULL.

9) YoY Sales Growth (%)

- **Definition:** Year-over-year percent change vs the same week number in prior year.

- **Formula:** `(sales_this_week - sales_same_week_last_year) / sales_same_week_last_year`
- **Grain:** week_start_date
- **Implementation note:** Use `EXTRACT(ISOWEEK FROM date)` + `EXTRACT(YEAR FROM date)` matching, or a 52-week lag approach.
- **Edge cases:** Missing prior-year week → NULL.

10) Rolling 4-Week Sales (MA4)

- **Definition:** 4-week moving average of total sales.
- **Formula:** `AVG(total_sales) OVER (ORDER BY week ROWS BETWEEN 3 PRECEDING AND CURRENT ROW)`
- **Grain:** week_start_date
- **Notes:** Smooths noise; great for exec view.

11) Rolling 8-Week Trend Slope

- **Definition:** Simple trend indicator using last 8 weeks; can be slope or % change.
- **Option A (simple):** `(sales_current - sales_8_weeks_ago) / sales_8_weeks_ago`
- **Option B (slope):** linear regression slope in Python (more advanced).
- **Grain:** week_start_date
- **Notes:** Use for "at-risk" scoring.

12) Top Movers (Contribution to Change)

- **Definition:** Which stores/depts contributed most to ΔSales between two periods.
- **Formula:**
  - `delta = sales_period2 - sales_period1` by store/dept
  - sort by `ABS(delta)` or `delta`
- **Grain:** store_id or dept_id (for a chosen comparison window)
- **Notes:** This is your "why behind the numbers."

Holiday and seasonality

13) Holiday Sales ($)

- **Definition:** Total sales during holiday weeks.
- **Formula:** `SUM(Weekly_Sales) WHERE IsHoliday = TRUE`
- **Grain:** week_start_date (or period)

14) Non-Holiday Sales ($)

- **Definition:** Total sales during non-holiday weeks.
- **Formula:** `SUM(Weekly_Sales) WHERE IsHoliday = FALSE`

15) Holiday Lift (%)

- **Definition:** Percent lift of holiday-week sales compared to non-holiday baseline.
- **Formula:** `(avg_holiday_sales - avg_nonholiday_sales) / avg_nonholiday_sales`
- **Grain:** segment (store, dept, type) + period
- **Edge cases:** baseline = 0 → NULL
- **Notes:** Also compute by store type and dept for "sensitivity."

16) Holiday Contribution (% of Total)

- **Definition:** Fraction of sales that occurs during holiday weeks.
- **Formula:** `holiday_sales / total_sales`
- **Grain:** period

17) Holiday Sensitivity Ranking

- **Definition:** Rank stores/depts by Holiday Lift.
- **Formula:** `RANK() OVER (ORDER BY holiday_lift DESC)` within segment.
- **Notes:** Shows where holidays matter most.

Stability / volatility / risk

18) Volatility Index (CV)

- **Definition:** Coefficient of Variation over a rolling window (e.g., 8 weeks).

- **Formula:** `STDDEV(total_sales) / AVG(total_sales)` over rolling N weeks
- **Grain:** week_start_date (or store/dept + week)
- **Notes:** Higher = less predictable; a risk signal.

19) Sales Anomaly Flag

- **Definition:** Flags weeks where sales deviates strongly from recent history.
- **Formula (simple):**
  - compute rolling mean & stddev over N weeks
  - flag if `ABS(sales - mean) > 2 * stddev`
- **Grain:** week_start_date (or store/dept + week)
- **Notes:** Great for DataOps-like monitoring.

20) At-Risk Score (Store/Dept)

- **Definition:** Composite score to prioritize where attention is needed.
- **Suggested components (0–100 scaled):**
  - Trend decline (rolling 8-week % change)
  - High volatility (CV)
  - Recent negative WoW
  - Optional: holiday sensitivity mismatch (if holiday periods)
- **Example formula (conceptual):** `risk = 0.5*trend_decline_score + 0.3*volatility_score + 0.2*recent_drop_score`
- **Grain:** store_id or dept_id + current period
- **Notes:** Keep weights documented. This is your “senior signature.”

Data quality / governance checks (DataOps lite)

21) Freshness Check

- **Definition:** Confirms latest week exists and data is up-to-date.
- **Check:** `MAX(Date)` is within expected range (for static dataset, simulate).
- **Notes:** In real pipelines, this is critical.

22) Uniqueness Check (Primary Key)

- **Definition:** Ensure each store+dept+week has at most one row.
- **Check:** count duplicates of `(Store, Dept, Date)` should be 0.

23) Completeness Check (Critical fields)

- **Definition:** No nulls in `Store, Dept, Date, Weekly_Sales`.
- **Check:** null counts = 0 (or within acceptable threshold)

Naming conventions (recommended)

- Facts: `fct_...` (e.g., `fct_weekly_sales`)
- Dimensions: `dim_...` (e.g., `dim_store`)
- Curated views: `vw_...` (e.g., `vw_sales_kpis_weekly`)
- Metrics in BI: `metric_total_weekly_sales`, `metric_wow_growth`, etc.

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